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The Intersection of Art and Technology: Innovative Thinking and Practice in the Perspective of New Media

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Abstract

The swift advancement of science and technology has emerged as a pivotal catalyst for artistic innovation, with immersive art displays becoming increasingly prevalent as a favored mode of creative expression. This paper conducts a thorough analysis of the flow dynamics within immersive art display spaces, utilizing the eight-channel cardio fluid flow model as a foundation. By integrating the immersive communication effect model, this study further investigates the communicative aspects of these spaces. The implementation of human-computer interaction technologies not only enhances visual presentation but also facilitates the design of immersive art spaces through advanced scene modeling techniques. This design process includes optimizing the viewer's perspective by incorporating visual perception parameters and the grid saliency MS algorithm. Additionally, this research undertakes a quantitative evaluation of interactive applications at the nexus of art, science, and technology, focusing on audience demand, interactive experiences, and recognition. The findings reveal a significant increase in audience demand for digital media interaction, from 12.64% to 31.29%, and an impressive average tolerance score of 84.32 for immersive art spaces. Comments related to these spaces uniformly express positive recognition, underscoring the successful integration of science and technology in art displays. This integration, as demonstrated, fosters a diverse evolution of artistic expression forms.

Keywords: Heart flow model; Human-computer interaction; Scene modeling technology; Visual perception; Grid saliency; Art.

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1 Introduction

Today's digital information dissemination brings us unprecedented audiovisual experience. In the era of interconnected information, we can enjoy all effective messages in the most convenient way [1-3]. The development process of art is no exception, from the initial simple book interpretation illustrations evolved to today's expression of a unique form of artistic creation of colorful, hand-drawn illustrations that has a deeper level of artistic value [4-6]. Throughout the brief history of the development of art, it is not difficult to see that each period of art style is closely related to the contemporary cultural and economic status [7-9]. Therefore, in the 21st century, how to maintain the purity and traditional appearance of hand-drawn illustration art, and at the same time make it able to play its maximum value in today's society with diversified information is a proposition worthy of continuous thinking by many practitioners [10-12]. The digital new media bring us a new way of accepting information. At the same time, science and technology communication modes have also produced a certain degree of change in our mode of thinking. The traditional in the new trend of thought under the impact of the new trend must be new and change, which is also in the creation of us at the same time to emphasize the innovation of this concept, and this concept is consistent with the future development of all creative industries and positioning requirements [13-15]. Therefore, we should not only pay attention to the fresh communication methods and carriers in the new media era but also explore the innovative soul of hand-painted illustrations from the inside [16-18]. With the updating of knowledge, how can we maintain the artistic charm of hand-drawn illustration and at the same time be able to accomplish a higher degree of cultural transmission through creation, which is also a mission of our creative research [19-20].

Based on the eight-channel flow model of cardio fluid, this paper explores the specific flow history of digital media art. It analyzes the richness of its communication content in combination with the art space communication effect model. The technical support for human-computer interaction required for immersive art display space is sorted out, and in order to realize the visual expression of art display space, scene modeling technology is used for the design of immersive art display space. The visual perception features are introduced into the immersive scene, combined with the grid saliency MS algorithm to select the optimal viewpoint of the immersive scene, which in turn improves the visual expression of the immersive art display space. For the cross-application of art and technology, this paper analyzes the audience situation, interactive experience, and audience recognition degree of the art display space from three aspects in order to provide a new reference direction for the integration of art and technology.

2 Design framework for immersive art exhibition spaces

Art exhibition space is a place for inheriting human civilization and spreading scientific knowledge and culture. In the background of the digital media era, the exhibition space tends to be more diversified in the form of information dissemination. The design development trend of modern art exhibition space is closely connected with the development of the economic environment, science and technology level, and humanistic needs of today's society, and the transformation of a service-oriented society to an experience-oriented society makes art creation and dissemination appear in a new form of exhibition. The application of science and technology for the dissemination of art and culture to provide new inspiration and methods, new media technology, and scientific and technological innovation progress for modern art exhibition space design has brought technical support and enrichment, the development of many new forms of communication media.

2.1 The History of Digital Media Art

2.1.1 Eight-channel flow model of cardiac flow

Mainstreaming, the state in which a person is completely immersed in an activity, oblivious to the existence of anything else, is an experience that brings great joy in itself and makes a person willing to go to great lengths. When a person is engaged in a challenging task, they enter a state of total concentration, oblivion, and immersion, which can bring great satisfaction and enjoyment. In immersive art spaces, the theory of mind flow is widely used, and the eight-channel flow model of mind flow is shown in Figure 1. By creating challenging artworks and exhibition spaces, artists attract the audience to enter the state of mind flow so that the audience can be immersed in it, forget about the surrounding environment, and completely devote themselves to the artworks and exhibition spaces.

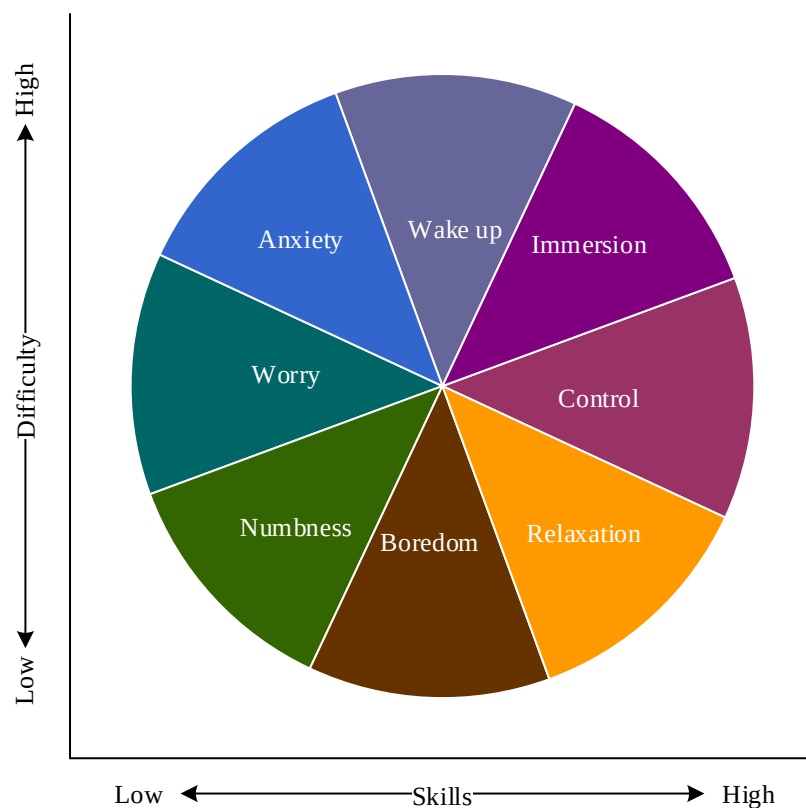


Figure 1. The flow model of the heart flow

Design in immersive art spaces often includes elements of sound, visuals, interaction, and performance. By combining these elements, a challenging and fascinating environment can be created that can stimulate the audience's curiosity and inquisitiveness to enter a state of mind flow. In this state, the audience can experience more deeply the beauty and value brought by artworks and exhibition space, which improves their art appreciation and aesthetic ability.

Therefore, the application of the theory of mind flow provides an effective way for immersive art spaces so that artists and audiences can communicate and exchange better and improve the audience's participation and the quality of the art experience. By better understanding the audience's needs and creating more challenging and attractive artworks and exhibition spaces, artists can promote the innovation and development of immersive art spaces.

2.1.2 Immersive Art Spatial Fluxions

Digital media art immersive scene design that is, in the context and scope of digital media art, to immersive design as a creative approach, scene design as a specific means of creation of digital media art, immersion is the result of this art form. Relying on digital media technology will be art through the computer language and the form of digital media to show is a science, art, and media, communication of the three crosses produced by the new art performance and application of the discipline. Based on the interactivity, hyperlinks, cross-media, and other characteristics of digital media technology, immersive art exhibition spaces can help the experiencer realize the real experience. The evolution of immersive art exhibition space combined with digital media technology has gone through four stages of development: the preparation period, the initial stage, the growth period, and the prosperity period, which enables it to debut on a large scale in the new media era.

- 1) Preparatory period (late 19th century) - the romantic language of Madame Luce.
- 2) Inception period (1950s to 1960s) - the very different reflections of scientists and artists.
- 3) Growth period (1970s to 1980s) - the rapid spread of mass media.
- 4) Prosperity period (1990s onwards) - the full power of the Internet.

The immersive art display space supported by digital media technology is a combination of science and technology, visual art, and media culture, and the design of immersive art display space with its scope and background is even more derived from the theory of positive psychology's mind flow. The rapid development of science and technology has made it one of the important factors in promoting artistic innovation, such as making artworks not only by virtue of a single visual sense while mobilizing the viewer's multi-sensory system to enter the world of works to enhance the immersive experience of artworks. "Immersion" is a special phenomenon in human spiritual ecology. New media art takes the creation of "immersion" as one of its pursuit goals, and new media art theory also takes "immersion" as an important category. The theory of new media art also regards "immersion" as an important category.

2.2 Spatial richness of immersive art

2.2.1 Modeling the effects of immersive communication

For the immersive art exhibition space designed by the combination of digital technology and art in the new media era, different audiences' perceptions of contemporary art vary greatly. Relying on the immersive art exhibition space can help the audience express their views on art more clearly, explicitly, and individually, not limited to the level of information but closer to mobilization. The immersive communication effect model of different types of art audiences is shown in Figure 2, which divides art audiences into general audiences, enthusiasts, and professional audiences and mainly includes the performance of the audience's cognition, attitudes, emotions, and behaviors in many different dimensions.

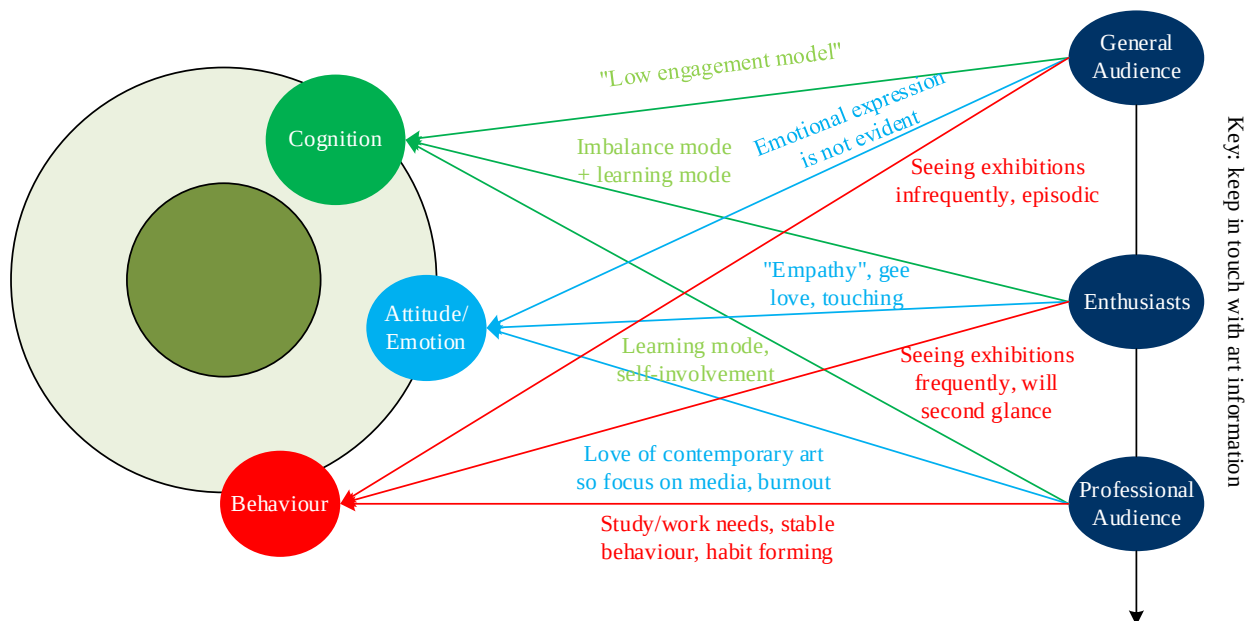


Figure 2. Immersive propagation effect model

To improve communication, it is necessary to maintain the audience's exposure to the art media message without interruption throughout the communication process. Maintaining and strengthening the audience's cognitive interest in art is the key so that they can learn again after the initial understanding of the exhibition information, rather than interrupting the contact with art in a single communication. Immersive art exhibition spaces need to lower the threshold to give audiences more opportunities to participate and experience and deepen the connection with the general public's life so that audiences can enhance their knowledge of art through the learning mode.

2.2.2 Communication content of the art space

- 1) As a diversified form of development of art exhibition, immersive art exhibition space still possesses modeling and expressiveness to show its unique art style, which is also an important reason to bring a sense of immersion. It uses light and shadow, color, sound, and other symbolic elements to complete the modeling and express the immersive aesthetic feeling, making people experience the immersive aesthetic feeling.
- 2) Artists use virtual reality technology, multi-channel interactive technology, or mechanical numerical control devices to create immersive environments for users to experience. Digital cameras, infrared sensors, remote controls, and other information collection tools capture the audience's language, expression, movement, or other body language and data analysis after the computer program processing, and then through the sound, image, music, light, digital video, synthesized animation and mechanical interactive devices and other ways to feedback to the audience relevant information.
- 3) The immersive art exhibition space will design each audience to become a character in the narrative according to the aesthetic characteristics, storyline, and the public's aesthetic and cognitive needs, centering on the audience and adding a sense of immersion. Immersive art exhibition space across the boundaries of time and space, realizing the fusion of reality and virtual reality, mobilizing the audience's multiple senses to build a sense of immersion.

Immersive art display spaces allow the audience's passive position to become active. People can interact with the artwork, resulting in a strong sense of participation and immersion.

3 Visual design of immersive art exhibition space

Nowadays, with the continuous progress of science and technology, a variety of digital media applications have been a new development, more and more immersive art exhibition space, the use of new media forms of publicity and explanation of exhibits, to a large extent, can reflect the interactive development of modern information technology, create the atmosphere of the space. The rise of digital media technology has greatly improved the immersive art exhibition space design, with greater flexibility and vividness for visitors during immersion tests.

3.1 Technical support for art exhibition space

3.1.1 Visual presentation of human-computer interaction

In order to better present visual design in human-computer interaction, it is necessary to have a better understanding of the composition and mode of human-computer interaction. In terms of the composition of HCI, existing studies have summarized the basis of interactive activities into six elements, which are feedback, control, creativity, productivity, communication, and adaptability. So, these six elements are used as a scale to measure interactivity. The higher the scale, the stronger the interactivity. In the design, we need to understand the relationship between the six elements and grasp the interaction mode of human-computer interaction between the display interface and the media space. The interaction between human beings and media indirectly through objects is called the HPOM interaction mode.

The HPOM interaction mode applies to the interaction based on the online end, which refers to the basic architectural mode in that the object itself possesses computing power or connects to the external computing environment to influence the composition of the media space. Users can interact with the media space through the operation of object tools, such as cell phones, tablets, computers, and other objects. The audience can browse the information of the exhibition and even visit the online virtual space through their cell phone. And can also interact with the operation interface through the cell phone and can choose their favorite graphics and colors in the interface to create the related exhibition theme products. Online instructions can be used to synchronize relevant creations with physical production, taking advantage of the advantages of the network medium for consumers to enjoy interactive fun.

3.1.2 Reserve support for required technology

Before the visual image design of the immersive art display space is launched, first of all, according to the attributes and characteristics of the digital media of the art display space, the required digital media technology should be prepared in the preliminary preparation of the visual image design, so as to better provide support for the design, presentation, and promotion of the visual image. The technical support required in the design of immersive art exhibition space is shown in Figure 3. In terms of dynamization and interactivity, the presentation of visual graphics is based on supporting webpage design, WAP design, UI design, and so on. In terms of scenarios and extensibility, the presentation of visual graphics relies on supporting AR/VR technology, big data cloud computing, network programming, and other aspects. The software involved in visual image design, in addition to the basic graphic design software, also needs to participate in the support of DW, AE, PR, C4D, 3Dmax, Easy Show, VirtualBox, and other software.

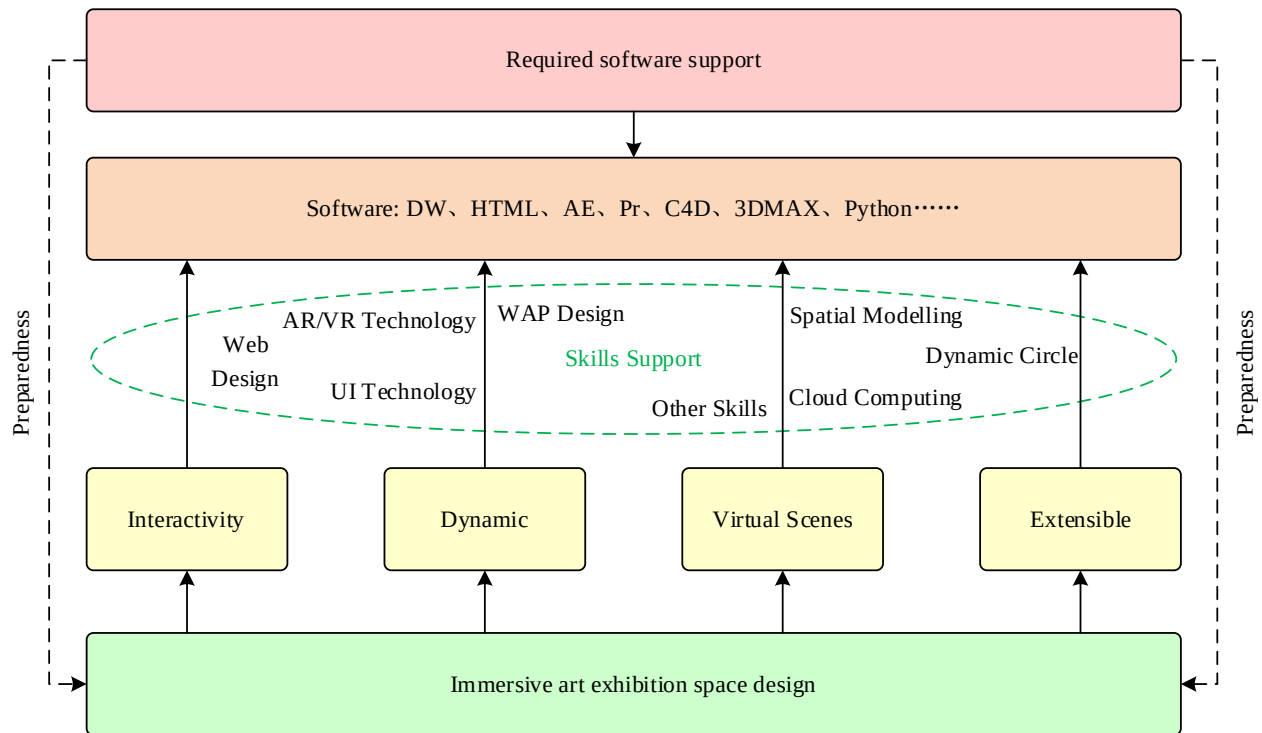


Figure 3. Reserve support for the required technologies

Relying on various technologies to enhance the design of immersive art display space, including the scene, visual image, human-computer interaction, and other dimensions, the use of science and technology, and the depth of the integration of art display to enhance the immersive sense of art display space and the sense of realism, to fully meet the audience's artistic experience.

3.2 Visual expression of art display space

3.2.1 Immersive Scene Modeling Techniques

Virtual reality technology is a technology that builds a virtual scene to simulate the real environment, which can be combined with sensors and other devices to complete human-computer interaction so that users can experience the advantages of virtual reality technology from visual, auditory, tactile, and other aspects. The virtual reality system realized by using virtual reality technology has three basic features, namely immersion, interactivity and multi-sensory, and the architecture of the virtual reality system designed based on these three features is shown in Figure 4. Developers first use 3D modeling, 3D animation and other technologies to create a good virtual scene, and then through human-computer interaction technology to connect hardware devices and complete the command operation, and finally complete the virtual reality system. Users can control objects in the virtual reality environment through sensors to complete human-computer interaction.

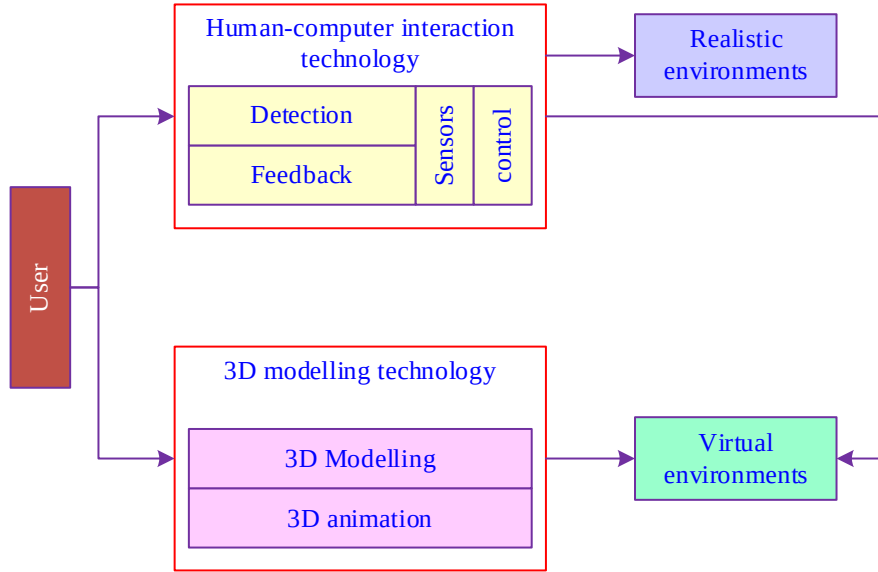


Figure 4. Immersive scenario modeling technique

Three-dimensional scene modeling needs to give a realistic three-dimensional immersion so that users with VR glasses in the three-dimensional modeling scene can have a sense of immersion. In order to achieve this goal, you need to take some environmental perspective in the real world for reference so as to build a better model in the virtual reality scene.

3.2.2 Visual perception of immersive scenes

After the immersive art display space scene design is completed, the metric method of each primitive feature information in the scene determines the speech sequence of the art display space model, which is a key factor affecting the quality of model simplification. In the model, feature information mainly includes geometric length, area, volume, shape (curvature), color, texture, lighting (normal vector), etc. However, most previous model simplification methods use the above feature information as a separate metric operator. Both geometric feature metrics and visual features should be included in the metric function of model simplification methods. Therefore, this paper mainly introduces metrics for visual features in immersive scenes, so as to enhance the user experience in immersive art display space scenes.

Visual feature information is the visually sensitive area of the human eye, i.e., the part of the object with large curvature (boundary, corner, crease). Visual feature information is usually measured using discrete curvature, calculated as:

$$Curv(v) = \sqrt{2H(v)^2 - K(v)} \quad (1)$$

$$H(v) = \frac{\sum_{e \in E_v} m(e)}{\frac{1}{3} \sum_{t \in T_v} a_t} \quad (2)$$

$$K(v) = \frac{s - \sum_{t \in T_v} \phi_t}{\frac{1}{3} \sum_{t \in T_v} a_t} \quad (3)$$

$$m(e) = \begin{cases} \gamma, & \text{if is convex} \\ 0, & \text{if is plane} \\ -\gamma, & \text{if is concave} \end{cases} \quad (4)$$

Where $H(v)$ is the mean curvature of vertex v , $K(v)$ is the Gaussian curvature of vertex v , $m(e)$ is the curvature function of edge e , a_t is the area of triangle t , γ is the normal angle between the planes adjacent to edge e , whose value is related to the associated planar attribute of the edge, ϕ_t is the angle of the set of the aircraft related with vertex v at vertex v , and s is the angular variable ($s = 2\pi$ if vertex v is an inner vertex and $s = \pi$ if vertex v is a boundary vertex).

The discrete curvature can be a better measure of the visual feature information of the model, but the computational terms of the equations of Eqs. (1) to (4) involve the area of the triangle associated with vertex v , the normal angle of the plane associated with edge e , and the angle at vertex v , and the arithmetic involves squares, roots, division, addition, subtraction, etc., which are complicated and computationally intensive and affect the speed of computation to a certain extent. In mathematics, the relationship between curvature and deflection is usually expressed as:

$$Curv(x) = \varphi(x)T(x) \quad (5)$$

Where x is the variable, $Curv(x)$ is the x curvature, $\varphi(x)$ is a function of x , and $T(x)$ is the x deflection.

The deflection equation for the vertex is expressed as:

$$T(v) = \frac{\sum_{v_i \in \text{planes}(v)} \|B_{v_i} - B_v\|}{\sum_{v_i \in \text{planes}(v)} \|v_i - v\|} \quad (6)$$

$$B_v = \frac{v \sum_{v_i \in \text{planes}(v)} \varepsilon_v n_{v_i}}{\left\| \sum_{v_i \in \text{planes}(v)} \varepsilon_v n_{v_i} \right\|} \quad (7)$$

Where $T(v)$ is the deflection of vertex v , $v \in V$, B_v are the subnormals of v , v is the moderating factor, ε_v is usually the area, $\text{planes}(v) = \{v_1, v_2, \dots, v_k\} \in V$ is the set of vertices in the 1-loop neighborhood of vertex v , v_i is the neighborhood vertices of vertex v , $v_i \in \{v_1, v_2, \dots, v_n\}$, n_{v_i} are the normal vectors of vertex v_i , and B_{v_i} is the subnormal of vertex v_i .

The deflection calculation term of Eq. (7) involves only the vertex normal vector and the triangle area, and the operations addition, subtraction, and division, which are relatively simple to compute, so the deflection rate is used to approximate the visual characteristics of the metric model. Since the computation of ε_v is relatively time-consuming, in order to reduce the amount of computation, a new computation is used to estimate the subnormal vector of the model vertices, and Eq. (7) can be expressed as:

$$B_v = v \sum_{v_i \in \text{planes}(v)} h_{v_i} n_{v_i} \quad (8)$$

$$h_{v_i} = \begin{cases} (n_v \cdot n_{v_i} - \varepsilon)^2 & \text{If } n_v \cdot n_{v_i} > \varepsilon \\ 0 & \text{If } n_v \cdot n_{v_i} \leq \varepsilon \end{cases} \quad (9)$$

Where h_{v_i} is the adjustment factor for vertex v_i and ε is the customization threshold.

The deflection rate indicates the distortion of the primitive. With the same value of deflection rate, the worse the triangle shape is, the worse the quality of this primitive is, and it needs to be prioritized to be removed. Therefore, the triangle shape factor weighted deflection is used to approximate the visual feature metric operator of the multi-feature metric operator. The minimum domain of action of the shape factor of a triangle is a triangle, and for computational simplicity, the deflection rate formula of an edge is notated as:

$$T(e_{ij}) = T(v_i) + T(v_j) \quad (10)$$

The visual feature metric operator for primitives consists of deflections weighted by triangular shape factors, which are computed as follows:

$$CB = \varsigma_{length} \cdot T(e_{ij}) \quad (11)$$

Where ς_{length} is the triangle shape factor, e_{ij} is the edge formed by vertices v_i and v_j , and $T(e_{ij})$ is the deflection of edge e_{ij} .

3.2.3 Intelligent selection of optimal viewpoints for scenes

In order to effectively obtain the optimal viewpoints in the 3D scene, the MS algorithm of grid saliency is proposed in existing research, which provides a new perspective for model area characterization. Its specific approach is as follows:

Definition $N(v, \sigma)$ is the set of vertices on the mesh whose distance from vertex v is less than σ , i.e., $N(v, \sigma) = \{x \mid \|x - v\| < \sigma\}$, x are the mesh vertices, assuming that the mean curvature of vertex v is $\zeta(v)$, and the Gaussian-weighted mean curvature at the vertices is given by Eq:

$$G(\zeta(v), \sigma) = \frac{\sum_{v \in N(v, 2\sigma)} \zeta(v) \exp\left[-\|x-v\|^2 / (2\sigma^2)\right]}{\sum_{v \in N(v, 2\sigma)} \exp\left[-\|x-v\|^2 / (2\sigma^2)\right]} \quad (12)$$

Where $\|v\|$ denotes the Euclidean norm of vector v . Define the significance value of vertex v under scale σ as:

$$\zeta(v)|_{\sigma} = |G(\zeta(v), \sigma) - G(\zeta(v), 2\sigma)| \quad (13)$$

To compute the mesh saliency at multiple scales, define the saliency of a vertex at scale i as:

$$\zeta(v)|_{\sigma_i} = |G(\zeta(v), \sigma_i) - G(\zeta(v), 2\sigma_i)| \quad (14)$$

Five scales are defined in MS $\sigma_i \in \{2\varepsilon, 3\varepsilon, 4\varepsilon, 5\varepsilon, 6\varepsilon\}$, ε as 0.3% of the diagonal distance of the enclosing box of the grid model. The significance values of the vertices at each of the five scales are calculated and normalized, and finally, the final significance value of vertex v is obtained by linear superposition:

$$\zeta(v) = \sum_{i=1}^5 S(\zeta(v)|_{\sigma_i}) \quad (15)$$

Where S is the nonlinear normalization function. In this paper, we take the model triangular facet as the object of calculation, extend the vertex saliency, define the saliency of the mesh model triangular facet T as the average of the saliency values of the three vertices of this triangular facet, and the calculation formula is as follows:

$$\zeta(T_i) = \frac{1}{3} \sum_{v \in T_i} \zeta(v) \quad (16)$$

Under a certain viewpoint, the larger the projected area of the triangular facet contributes more to the amount of information contained in that viewpoint. In order to reflect the effect of the perspective projection on the projected area of the triangular aspect, the above equation is modified by adopting the angle β_{TV} between the normal vector of the triangular facet N_T and the line-of-sight vector of the viewpoint N_V as a regulating parameter and the improved significance of T is defined as:

$$\zeta'(T_i) = w(T_i) \times \frac{1}{3} \sum_{v \in T_i} \zeta(v) \quad (17)$$

Where $w(T) = \sqrt{\cos \beta_{TV}}$, denotes the projection weight of the triangular facet T on the view plane.

In order to solve the problem of focusing too much attention on the local high-feature region and ignoring the global feature distribution, this paper introduces the concept of information entropy. Define the surface saliency of the mesh model under a certain viewpoint is expressed as:

$$S_E(T, Vp) = - \sum_{i=1}^{N_i} \frac{\zeta'(T_i)}{\sum_{T \in T_r} \zeta'(T_i)} \log \frac{\zeta'(T_i)}{\sum_{T \in T_r} \zeta'(T_i)} \quad (18)$$

Where $\Sigma_{T \in V_T}$ represents the set of triangular facets that are visible at the point of view. The visibility of a triangular aspect to the point of view is determined by whether all three vertices of the triangle are visible, and the visibility of the vertices can be determined by comparing the actual depth of the vertices to the value of the depth cache.

For the selection of the optimal viewpoint in the virtual scene, the MS algorithm is used to first calculate the mean curvature of the vertex, and the Gaussian-weighted mean curvature of the vertex under multiple scales is calculated by defining multiple scales $\sigma, 2\sigma, \dots, n\sigma$, respectively. The Gaussian-weighted mean curvature under the adjacent scales is used as the saliency value of the vertex under scale σ . Finally, the final saliency value of the vertex is obtained by nonlinearly normalizing the saliency values under different scales to get the final saliency value of the vertex.

4 Immersive art exhibition space application practice

The era of rapid change in science and technology is also affecting people's aesthetic interests and spiritual needs. The widespread use of new media technologies has had a significant impact on people's lifestyles. In art viewing activities, people are increasingly paying attention to the quality of the experience and spiritual satisfaction that the exhibition process brings. Therefore, combining new media and digital technology to create an immersive art exhibition space full of participation and experience provides a new direction for art development and art exhibition design.

4.1 Audience Analysis of Art Display Space

4.1.1 Access to information

Immersive art display space is designed to better help the audience understand the development of art in the new media era. The audience in the immersive art display space tour, the audience is mainly through what channels to understand to get into the immersive art display space of the relevant information is worth studying. The introduction of new media technology makes the communication channels of immersive art exhibition space more diversified. This paper adopts the questionnaire star to design the relevant questionnaires through the distribution of questionnaires to understand the audience to obtain the viewing channels of immersive art exhibition space. Audience access to information channels is shown in Figure 5.

Benefiting from the development of new media technology, more than 56.42% of the audience mainly obtains relevant information through the official WeChat public number and small program that carries out the immersive art exhibition space, 30.25% of the audience obtains relevant information from short video platforms such as Jittery, Shutterbug and so on, and 28.91% of the audience comes from the introduction of sharing based on the social platforms such as Xiaohongshu and Weibo. Most of the audience mainly learned about the immersive art display space through new media such as WeChat, Weibo, Xiaohongshu, Jieyin, Shutterhands, etc. And a small portion of the audience's attention was also attracted through traditional media such as newspapers and magazines, books, TV advertisements, and outdoor advertisements. Therefore, the access to immersive art exhibition space is mainly based on various all-media platforms supported by new media technology, which helps the

immersive art exhibition space to be more intuitively displayed in front of the audience and provides a diversified way for the audience to understand the immersive art exhibition space.

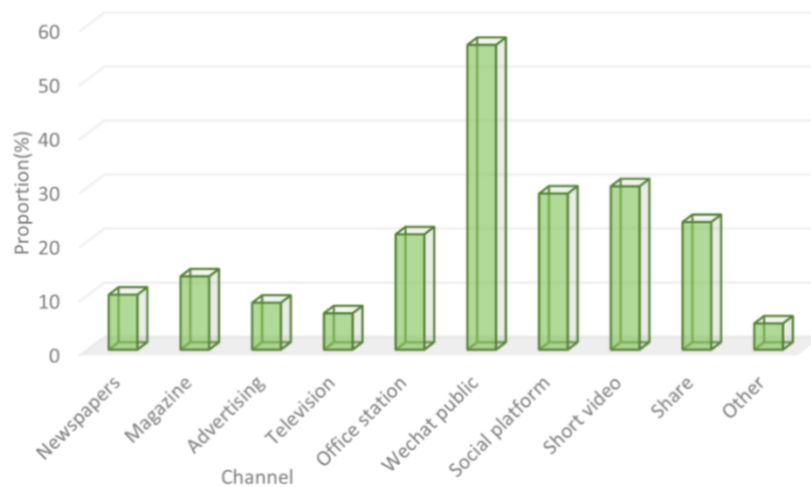


Figure 5. Audience access to information

4.1.2 Analysis of User Needs

The purpose of the immersive art display space is to help the audience better understand the development of art under the new media technology support. For the presentation of art display information, it is necessary to fully understand the user's needs in order to better give the art display space a deeper sense of immersion. As a result, this paper obtains the audience's demand for the presentation of immersive art display space information between 2015 and 2023 to sort out. It obtains the user's demand for the presentation of art display information, as shown in Figure 6.

Between 2015 and 2023, the audience's presentation of exhibit information for immersive art display spaces mainly focused on four types: textual presentation, scientific illustration, video narration, and digital media interaction, and this will increase over time. Audience demand for textual forms of art exhibit information presentation gradually decreased from 26.46% in 2015 to 11.94% in 2023, while audience demand for digital media interaction increased from 12.64% to 31.29%, an increase of 18.65 percentage points. During this period, science illustration and video narration fluctuate between 15% and 25%. The development of new media technology has led to a significant change in the audience's demand for immersive art display space display methods. The audience wants to show the relevant information of artworks through more interactive digital media methods to help the audience realize an immersive experience with more realistic scenes and to bring a better interactive experience to the audience.

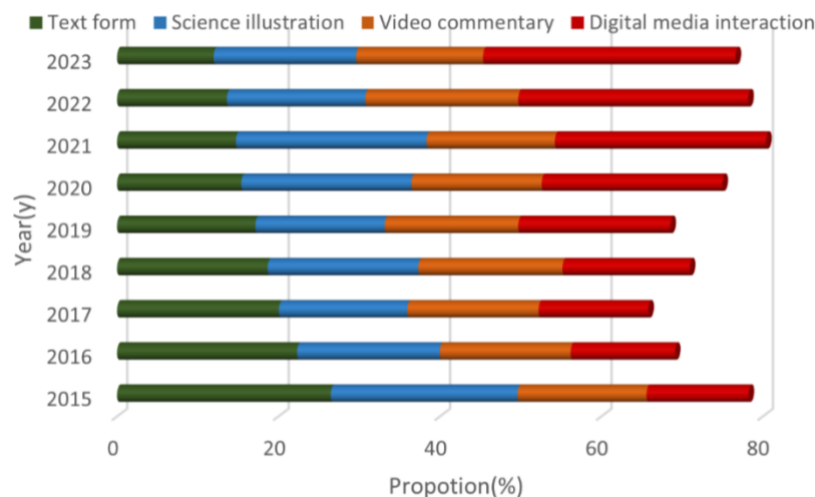


Figure 6. Presentation mode requirements

4.2 Interactive experience and audience identification

4.2.1 Audience Interaction Experience

In order to understand the degree of the audience's interactive experience of the immersive art display space constructed based on immersive scene modeling technology, this paper investigates the indicators of participation, pleasure, impression, and tolerance. Mainly from the ticketing service (A), information guide (B), activity design (C), equipment integration (D), information interpretation (E), cultural and creative products (F), spatial place (G), and service awareness (H) and other dimensions of the 100-point scale, the audience's interactive experience in the immersive art display space is shown in Figure 7.

From the audience's interactive experience score in the immersive art display space, it can be seen that the average value of its participation, pleasure, influence, and tolerance are all above 80 points. The average value of tolerance score for the immersive art display space is 84.32 points, in which the tolerance of cultural and creative products has reached 85.61 points. This reflects that the audience has a greater tolerance for the interactive experience of immersive art display space. In further optimizing the design of immersive art display space, it is necessary to keep abreast of the times by adopting more diversified science and technology and relying on the real-life simulation experience to increase the number of truly meaningful digital, intelligent, and operable interactive devices. Enhance the sense of integration of science and technology and art exhibits so as to better meet the audience's interactive expectations of the art display space and enhance the audience's understanding of immersion and participation in the art display space.

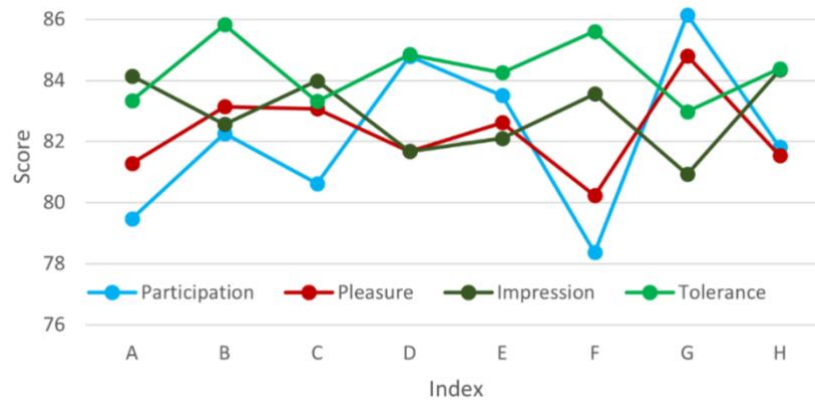


Figure 7. User Interaction Experience

4.2.2 Level of audience acceptance

With the continuous integration and development of new media and traditional media, the role and importance of comments in the all-media environment have gradually emerged, especially with the development of social media such as microblogging and WeChat, the user coverage is extremely high, and the influence is getting stronger and stronger, which makes many institutions, enterprises, and cultural industry organizations use WeChat public number to carry out marketing, and then to shape and spread their own media image. For the degree of audience recognition of immersive art display space, this paper selects the user comments of the relevant WeChat public number as the object of study, and through the statistical analysis of the content of the audience comments, attitude tendency, and combing of the comments, the audience's cognitive attitudes and degree of recognition of the immersive art display space are shown in Figure 8.

Based on the relevant comments received by the audience on WeChat, the following conclusions can be drawn:

- 1) The messages related to art literature and creativity in the comments of the immersive art display space show a growing trend from 2018 to 2022. Its growth from 315 to 506, whether it is the evaluation of the relevant art promotional films, online APPs, mini-games, or offline souvenirs, commemorative vouchers, and so on produced by the art display space, which is very high, and generally has a positive tendency. The aesthetics, creativity, culture, design, and practicality of the art and cultural creation series based on the art exhibition space have received wide praise from the audience.
- 2) Messages about the overall description of immersive art display space reached 166 in 2022, an increase of 69.39% from 2018. Most of the audience's overall description of the art display space uses words such as grounded, beautiful, three-dimensional and vivid, profound, vivid and fashionable, young and lovely, etc. The use of these words indicates that the image of the art display space in the audience's mind is no longer as high and cold as it used to be. Still, it has become more youthful, energetic, and popular with the collision of art history and modern technology to display a more humanized, visualized, and youthful media image.
- 3) The overall increase in art dynamic messages targeting immersive art display spaces is the largest, growing from 107 messages in 2018 to 224 messages in 2022, an increase of 1.09 times. Most of the content is shared by the audience for specific exhibitions and offline activities of the art exhibition space, which helps more audiences understand the development

of immersive art exhibition space. It can also be seen that the overall evaluation of the audience for the art exhibition space is positive. There is also the intention of “urging” the exhibition, which provides support for further promoting the development of immersive art exhibition space. This includes support for further promotion of the development of immersive art exhibition spaces.

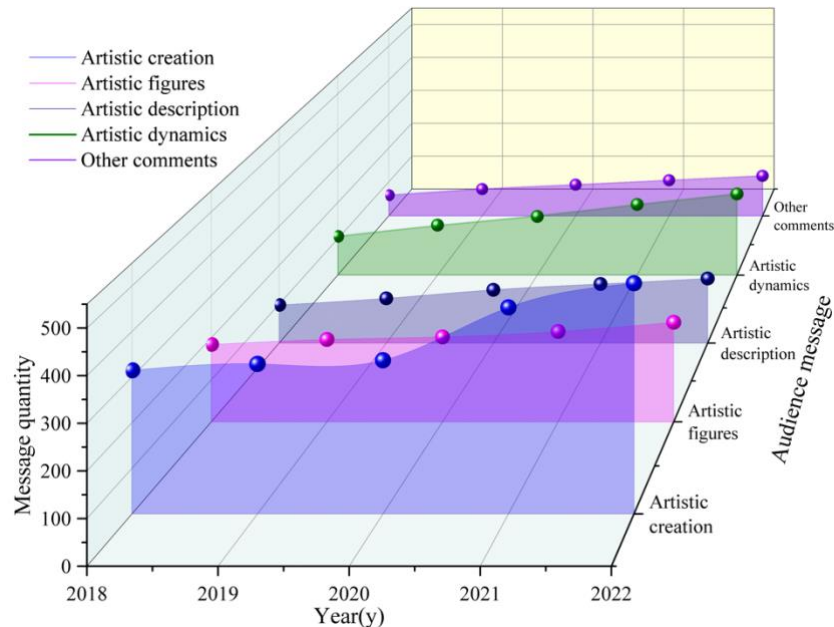


Figure 8. Audience recognition

5 Conclusion

This paper takes the flow history of digital media art as an entry point, analyzes the content richness of immersive art display space, combines human-computer interaction technology and immersive scene modeling technology for the construction of immersive art display space, and optimizes the visual expression of the art display space by using visual perception and the scene optimal viewpoint intelligent selection algorithm. To evaluate the practicability of immersive art display spaces, this paper validates the three dimensions of audience demand, interactive experience, and degree of recognition and draws the following conclusions:

- 1) The development of new media technology gives immersive art display space more diversified communication channels. 56.42% of the audience can obtain relevant information about the art display space through the official WeChat public number and small programs, and the audience's demand for digital media interaction has increased from 12.64% to 31.29%, an increase of 18.65 percentage points. The application of digital media technology promotes the development of immersive art display spaces, provides audiences with more diversified choices of information channels, and supports the communication of art.
- 2) The average values of audience participation, pleasure, influence, and tolerance in immersive art display space are all above 80 points, and the average value tolerance score for immersive art display space is 84.32 points. Relying on virtual reality technology and human-computer interaction technology, the immersive art display space can provide the audience with a more realistic art experience and help them understand the development of art in close proximity.

- 3) The audience's comments on the immersive art exhibition space show a greater positive tendency, and the comments show the degree of recognition of the immersive art exhibition space. Relying on science and technology can enhance the intimacy between the art display space and the audience, making the art display more humanized and youthful.

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